LINEAR PROGRAMMING: Simplex Method

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Maximize $z = 2x_1 + x_2,$

Subject to

$$4x_1 + 3x_2 \leq 12,$$

$$2x_1 + x_2 \leq 4,$$

$$x_1 + 2x_2 \leq 6,$$

$$x_1, x_2 \geq 0.$$

Standard form

$$4x_1 + 3x_2 + s_1 = 12$$

$$2x_1 + x_2 + s_2 = 4$$

$$x_1 + 2x_2 + s_3 = 6$$

$$x_1, x_2 \geq 0$$

s_1, s_2 and s_3 are slack variables

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution	Ratio
z	1	−2	−1	0	0	0	0	
s_1	0	4	3	1	0	0	12	3
s_2	0	2	1	0	1	0	4	2
s_3	0	1	2	0	0	1	6	6

Basic	z	x_1	x_2	s_1	s_2	s_3	Solution	Ratio
z	1	0	0	0	1	0	4	
s_1	0	0	1	1	−2	0	4	
x_1	0	1	1/2	0	1/2	0	2	
s_3	0	0	3/2	0	−1/2	1	4	

$$x_1 = 2, \quad x_2 = 0 \qquad z^* = 4$$

You can obtain other solution too by making x_2 basic.

Supposing in an LP problem, two optimal solutions $\boldsymbol{x}$ and $\boldsymbol{y}$ exist, prove that $(1-p)\boldsymbol{x}+p\boldsymbol{y}$ is also an optimum solution, if $0 \le p \le 1$.

Solution: The objective function of LP is $\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x}$, where $\boldsymbol{c}$ is the vector of coefficients. Let us assume that the optimum value of objective function is V. In view of $\boldsymbol{x}$ and $\boldsymbol{y}$ being the solution:

$$\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} = V \qquad (i)$$

$$\boldsymbol{c}^{\mathrm{T}}\boldsymbol{y} = V \qquad (ii)$$

Multiplying (i) by $(1-p)$ and (ii) by p and adding:

$$\boldsymbol{c}^{\mathrm{T}}\left\{(1-p)\boldsymbol{x}+p\boldsymbol{y}\right\} = (1-p)V + pV = V$$

This shows that $(1-p)\boldsymbol{x}+p\boldsymbol{y}$ is also a solution.

Note: Also, as the feasible space is convex in LP problem, then if $\boldsymbol{x}$ and $\boldsymbol{y}$ are feasible, then $(1-p)\boldsymbol{x}+p\boldsymbol{y}$ is also feasible.

Consider the following linear programming problem:

$$\text{Maximize } z = 2x + 4y$$

$$\text{subject to } x + 2y + S_1 = 5,$$

$$x + y + S_2 = 4,$$

$$x1\,, x2\,,\ S1\,,\ S2 \geq 0.$$

While solving this problem by simplex method, the following final tableau is obtained:

Basic	x	y	S1	S2	Solution
z	0	0	2	0	10
y	0	1	1	-1	1
x	1	0	-1	2	3

Carry out one more iteration and make $S2$ as a basic variable to obtain another optimum solution. Afterwards write a general expression for all alternative optima.

Sol: Given LPP: Maximize: $z = 2x + 4y$

Subject to: $x + 2y + S_1 = 5$; $x + y + S_2 = 4$; $x, y, S, S_2 \geq 0$

Basic	x	y	S_1	S_2	Solution
z	0	0	2	0	10
y	0	1	1	–1	1
x	1	0	–1	2	3

Step 1: Identifying Alternative Optima

Since S_2 has zero in the z-row, introducing it will not change $z_{max} = 10$. Choosing S_2 to enter and x to leave (by min

Step 2: Perform Pivot (S_2, x) = 2

New x-row: $R' = R_2 / 2$

New y-row: $R_1 = R_1 – (–1) R'$

New z-row: $Rz = Rz – 2 R'$

(By minimum ratio test → 3/2).

Step 3

Basic	x	y	S_1	S_2	Solution
z	0	0	1	–1	10
y	0	1	1.5	–0.5	2.5
S_2	0	0	-0.5	0.5	1.5

Step 4: General Expression for Optimal Solutions, first we got solution $(x, y)= (3,1)$, now we got $(0, 2.5)$

$$(x, y) = \lambda (3, 1)+ (1-\lambda) (0, 2.5)= (3\lambda, 2.5-1.5\lambda)\ 0 \leq \lambda \leq 1$$

This represents all possible optimal solutions.

Maximize $w = 2x_1 + 5x_2$

s.t

$$x_1 + 3x_2 \leq 40,$$
$$3x_1 + x_2 \leq 24,$$
$$x_1 + x_2 \leq 10$$
$$x_1 \geq 0, x_2 \geq 0.$$

Basic	x_1	x_2	x_3	x_4	x_5	Solution
w	-2	-5	0	0	0	0
x_3	1	3	1	0	0	40
x_4	3	1	0	1	0	24
x_5	1	1	0	0	1	10

Basic	x_1	x_2	x_3	x_4	x_5	Solution
w	3	0	0	0	5	50
x_3	-2	0	1	0	-3	10
x_4	2	0	0	1	-1	14
x_2	1	1	0	0	1	10

Maximize $w = 4y$

Subject to

$y \le 1,$

$y \le 3,$

y is unrestricted

Maximize $w = 4y^{+} - 4y^{-} + 0s_1 + 0s_2$

Subject to

$$y^{+} - y^{-} + s_1 = 1,$$
$$y^{+} - y^{-} + s_2 = 3,$$
$$y^{-}, y^{+}, s_1, s_2 \ge 0.$$

Basic	y^{+}	y^{-}	s_1	s_2	Solution
w	-4	4	0	0	0
s_1	1	-1	1	0	1
s_2	1	-1	0	1	3

Basic	y^+	y^-	s_1	s_2	Solution
w	0	0	4	0	4
y^+	1	-1	1	0	1
s_2	0	0	-1	1	2

From the tableau: $y^+ = 1$, and $y^- = 0$, which gives $y = y^+ - y^- = 1 - 0 = 1$

Therefore, the optimal solution is: $y = 1$. Maximum $w = 4 \times 1 + 4 \times 0 = 4$

Maximize $z = 2x + y$

subject to

$$x - y \leq 0,$$
$$2x + 3y \leq 30,$$
$$x, y \geq 0.$$

Basic	x	y	s_1	s_2	Solution
z	-2	-1	0	0	0
s_1	1	-1	1	0	0
s_2	2	3	0	1	30

Basic	x	y	s_1	s_2	Solution
z	0	-3	2	0	0
x	1	-1	1	0	0
s_2	0	5	-2	1	30

Basic	x	y	s_1	s_2	Solution
z	0	0	4/5	3/5	18
x	1	0	3/5	1/5	6
y	0	1	-2/5	1/5	6

Using MATLAB (LINPROG function) or any other code for solving LP problem, solve the following problems:

Maximize $z = 2A+B-3C+5D$

Subject to

$$A+2B+2C+4D \leq 40$$
$$2A-B+C+2D \leq 8$$
$$4A-2B+C-D \leq 10$$
$$A, B, C, D \geq 0$$

%% Use of linprog function to solve linear programming problems (Source: https://in.mathworks.com/help/optim/ug/linprog.html)

```
clc; clear; close all;

A = optimvar('A', 'LowerBound', 0);
B = optimvar('B', 'LowerBound', 0);
C = optimvar('C', 'LowerBound', 0);
D = optimvar('D', 'LowerBound', 0);
```

```
% Define Optimization Problem
prob = optimproblem ('Objective', 2*A + B - 3*C + 5*D,
'ObjectiveSense', 'max');

% Define Constraints (Flipping "≥" to "<=")
prob.Constraints.c1 = A + 2*B+2*C+4*D <= 40;
prob.Constraints.c2 =  2*A - B + C + 2*D <= 8;
prob.Constraints.c3 = 4*A-2*B+C-D <= 10;

% Convert Problem to Solver Form
problem = prob2struct(prob);

% Solve using linprog
[sol, fval, exitflag, output] = linprog(problem);

% Display Solution
disp('Optimal Solution:')
disp(table(['A'; 'B' ; 'C'; 'D'], sol, 'VariableNames', {'Variable', 'Value'}))
disp('Optimal Objective Value:')
disp(-fval)
```

Optimal solution found.

Optimal Solution:

Variable	Value
A	0
B	6
C	0
D	7

Optimal Objective Value:

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